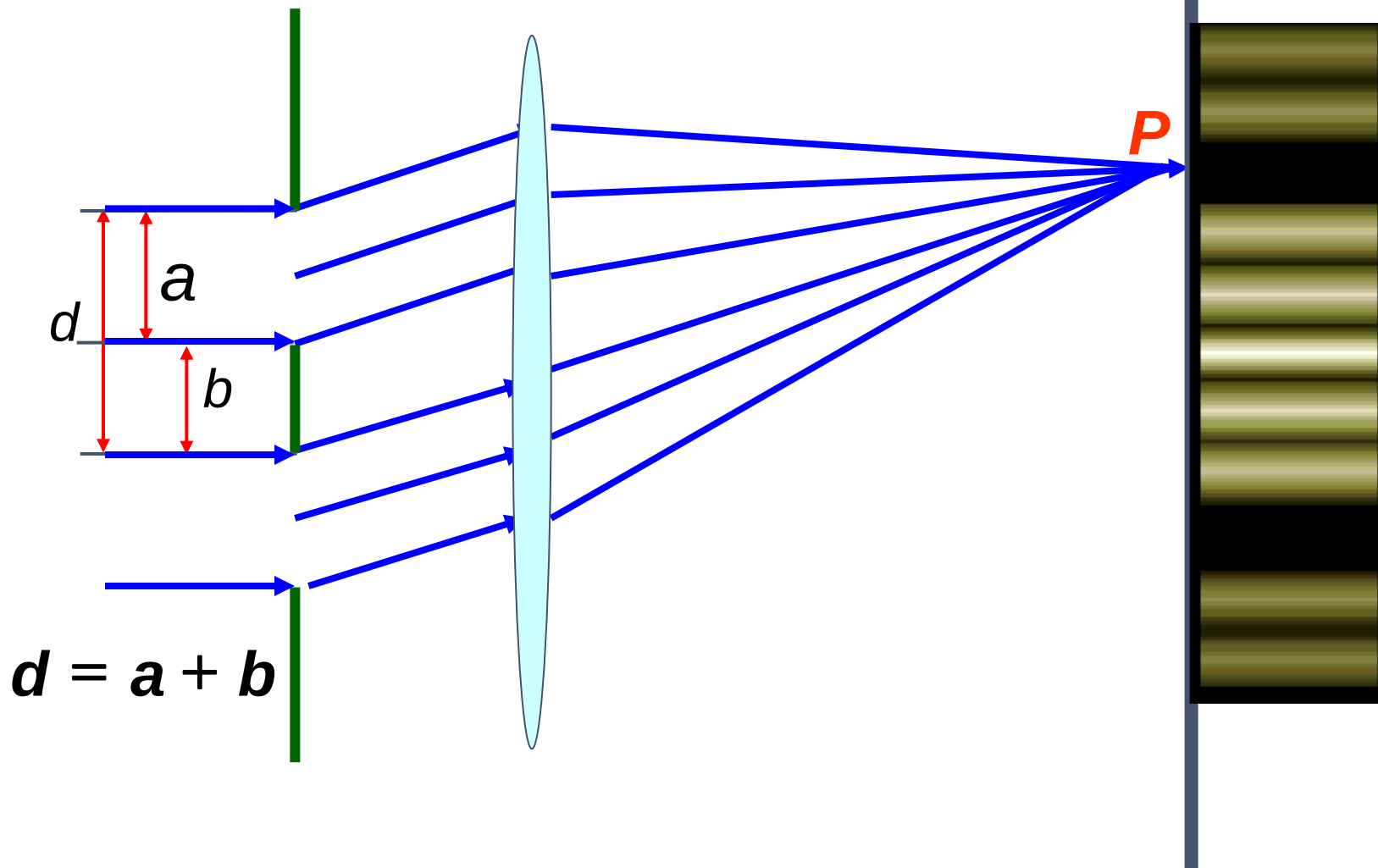


本节课作业

P93: 13-T20~T26

2. 双缝衍射现象

a : 缝宽, d : 缝间距



将双缝干涉与单缝衍射结合考虑

杨氏双缝干涉强度分布：

两条强度为 I_0 的光干涉

$$I_{\theta} = 4I_0 \cos^2 \beta$$

单缝衍射强度分布

$$I_{\theta \text{单衍}} = I_{m \text{单衍}} \left(\frac{\sin \alpha}{\alpha} \right)^2$$

双缝衍射的强度分布：

两束强度为 $I_{\theta \text{单衍}}$ 的光干涉

$$\alpha = \frac{\pi a \sin \theta}{\lambda}$$

$$I_{\theta \text{双衍}} = 4I_{m \text{单衍}} \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta$$

$$\beta = \frac{\pi d \sin \theta}{\lambda}$$

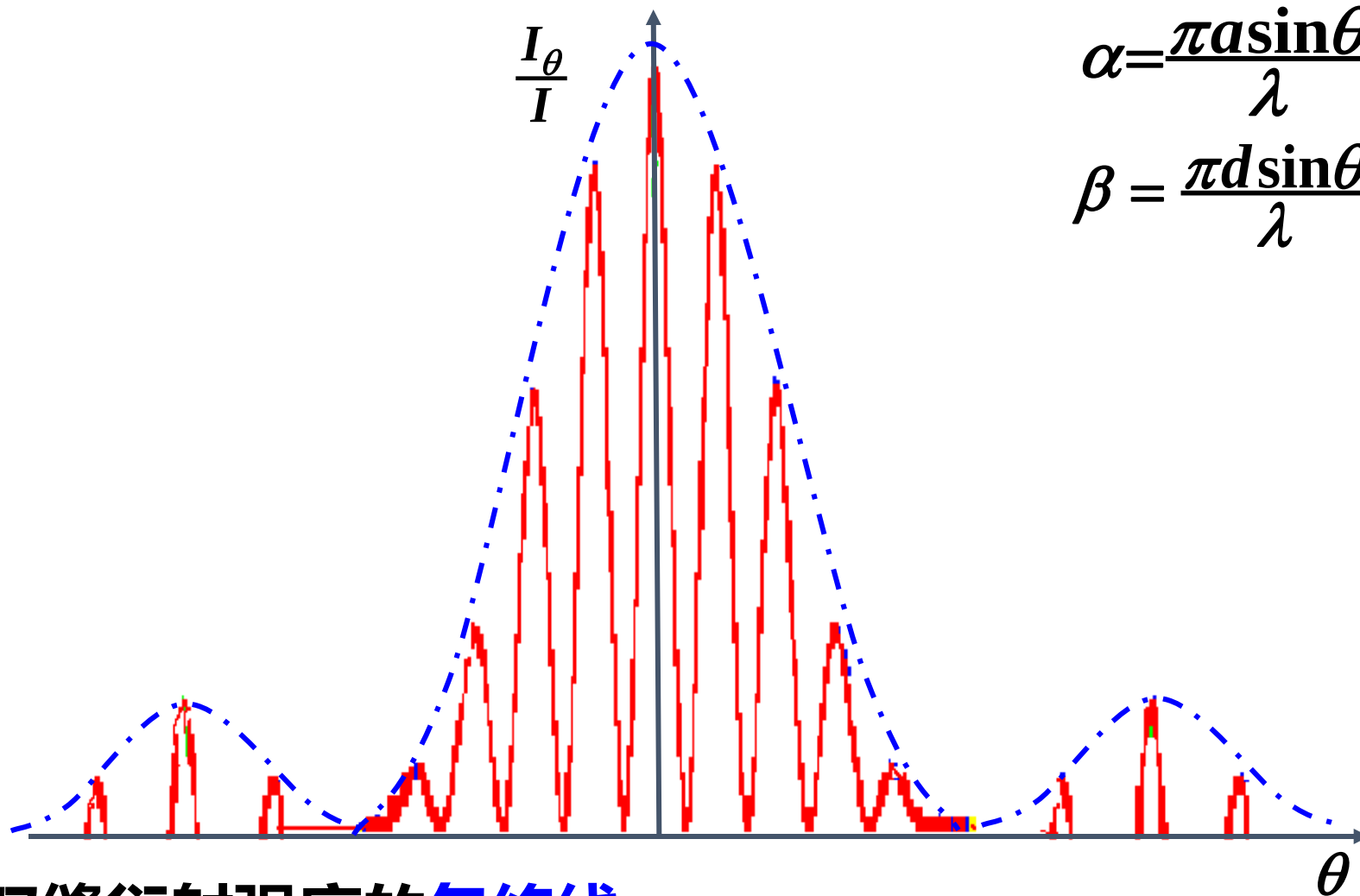
衍射因子 干涉因子

双缝衍射的强度曲线分布图

$$I_{\theta} = 4I_m \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta$$

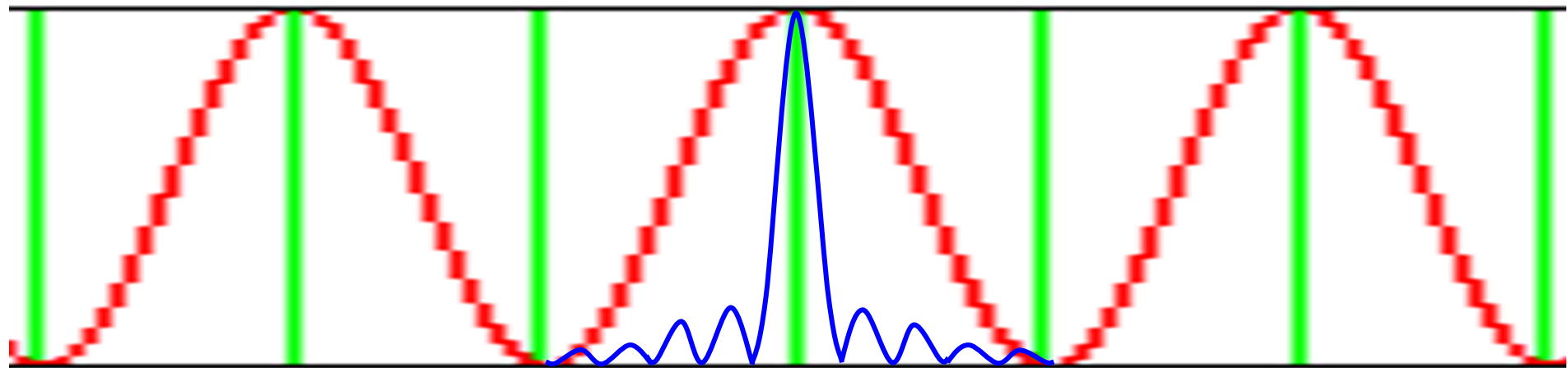
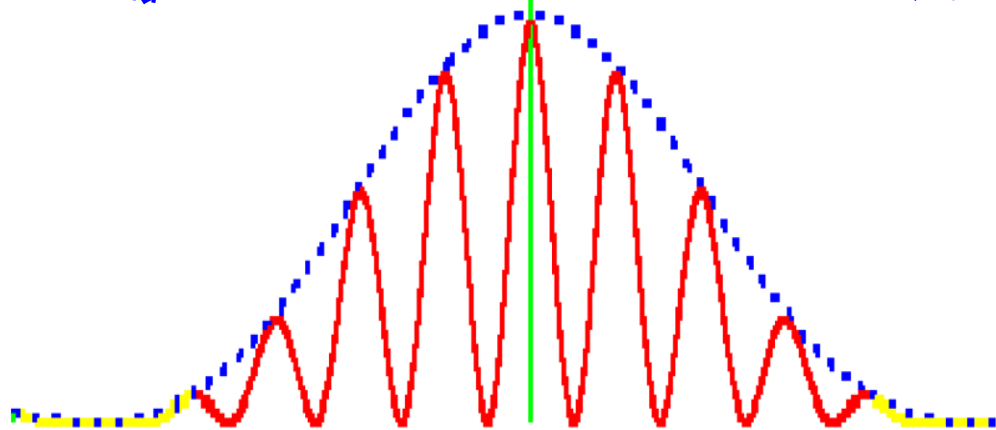
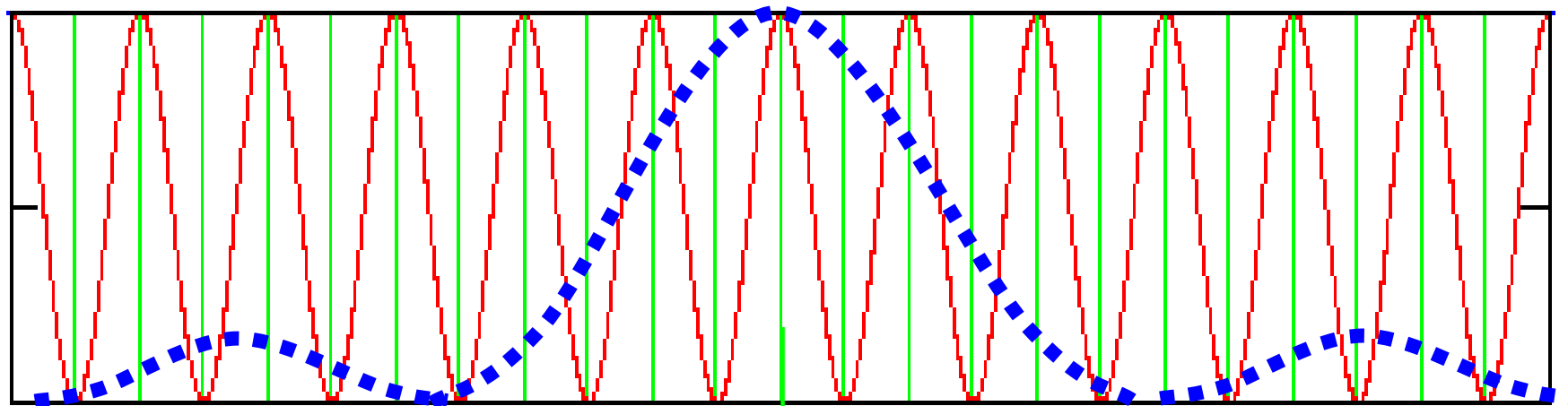
$$\alpha = \frac{\pi a \sin \theta}{\lambda}$$

$$\beta = \frac{\pi d \sin \theta}{\lambda}$$



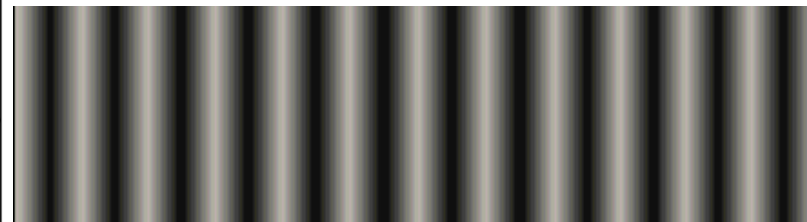
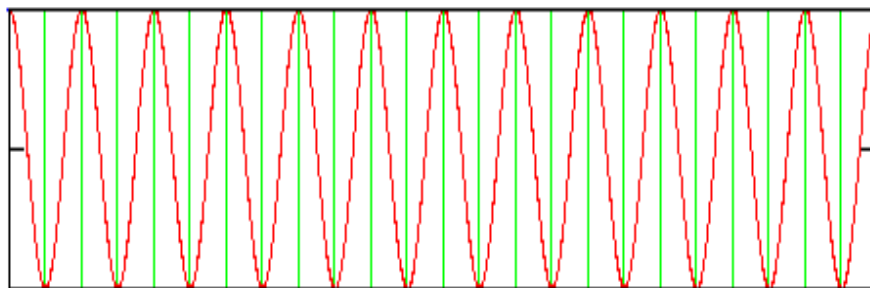
双缝衍射强度的**包络线**

——是单缝衍射强度分布



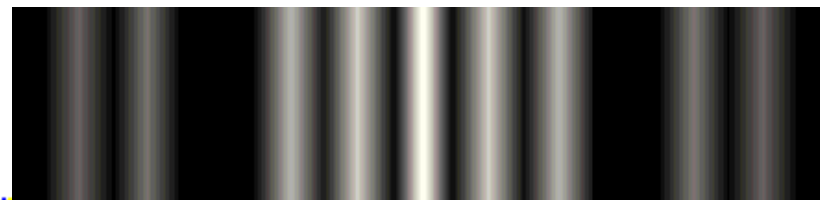
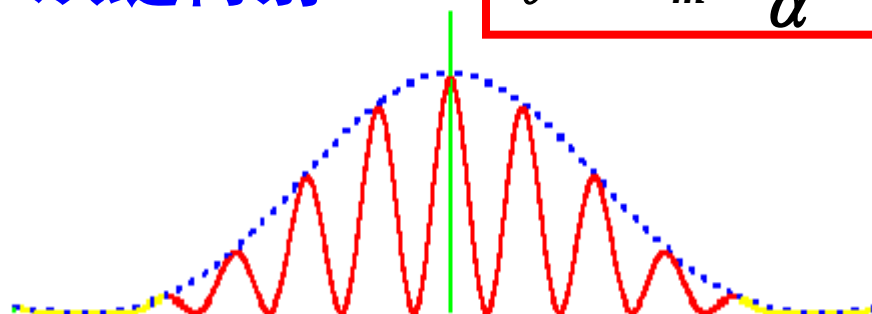
双缝干涉

$$I_{\theta} = 4I_0 \cos^2 \beta$$



双缝衍射

$$I_{\theta} = 4I_m \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta$$

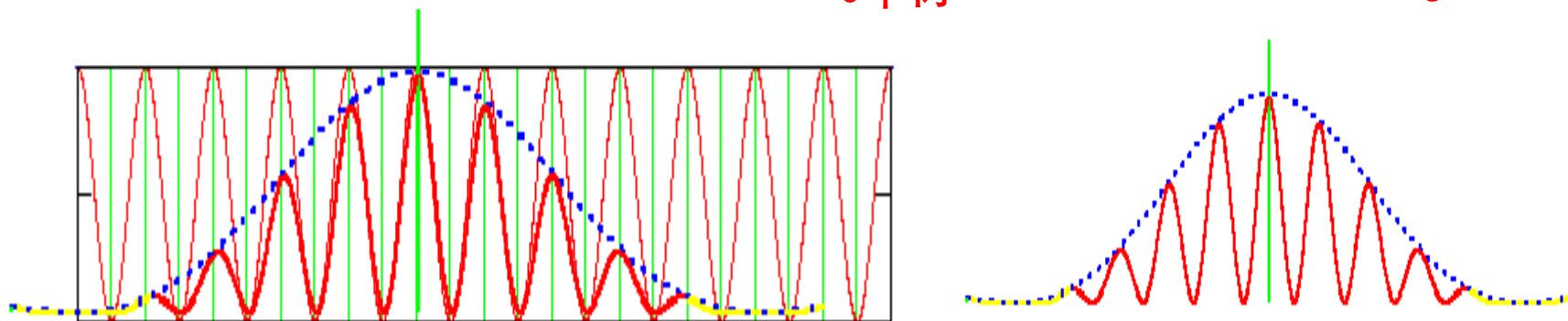


双缝干涉的条纹位置仍不变,只是强度受到单缝衍射的调制。

结论

1° 双缝衍射的强度曲线是单缝衍射强度对双缝干涉强度进行调制的结果。

这种调制表现在以变化的 $I_{\theta \text{单衍}}$ 代替了不变的 $4I_0$

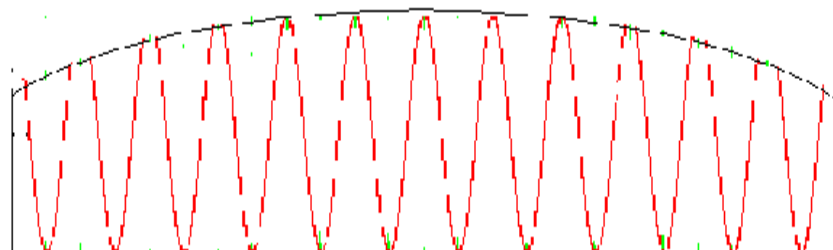


2° 在 $a \ll \lambda$ 时，双缝衍射的强度分布情况变为理想的杨氏双缝干涉的强度分布情况。

$$\alpha = \frac{\pi a \sin \theta}{\lambda} \Rightarrow 0$$

$$\Rightarrow I_{\theta} = I_m \cos^2 \beta$$

$$I_{\theta \text{双衍}} = I_{m \text{双衍}} \cos^2 \beta$$



4.双缝衍射明暗条纹的位置

极小值的位置

双缝干涉的极小(两束光位相相反)

$$d \sin \theta = (2k+1) \frac{\lambda}{2} \quad k = 0, \pm 1, \pm 2 \cdots \min(1)$$

单缝衍射的极小

(两束光迭加,强度为零)

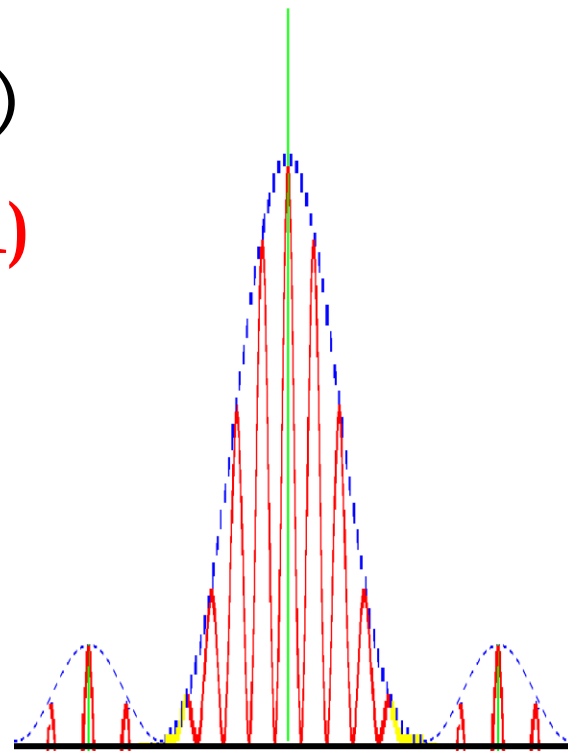
$$a \sin \theta = k' \lambda \quad k' = \pm 1, \pm 2 \cdots \min(2)$$

极大值的位置

双缝干涉的极大

$$d \sin \theta = k \lambda \quad k = 0, \pm 1, \pm 2 \cdots \max(3)$$

单缝衍射的极大在此无意义



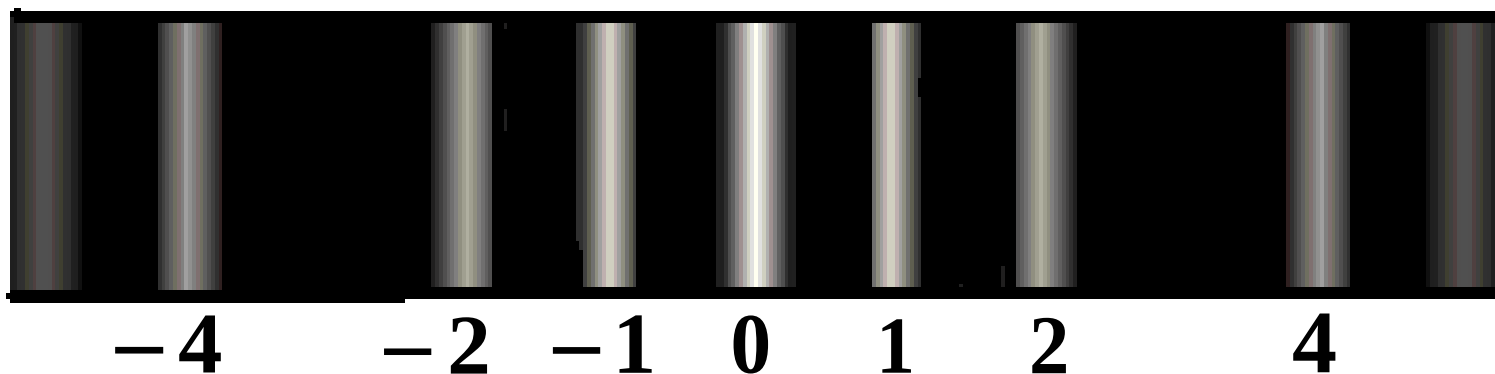
$$\left\{ \begin{array}{l} a \sin \theta = k' \lambda \quad k' = \pm 1, \pm 2, \dots \cdot \text{min} \dots (2) \\ d \sin \theta = k \lambda \quad k = 0, \pm 1, \pm 2, \dots \cdot \text{max} \dots (3) \end{array} \right.$$

讨论

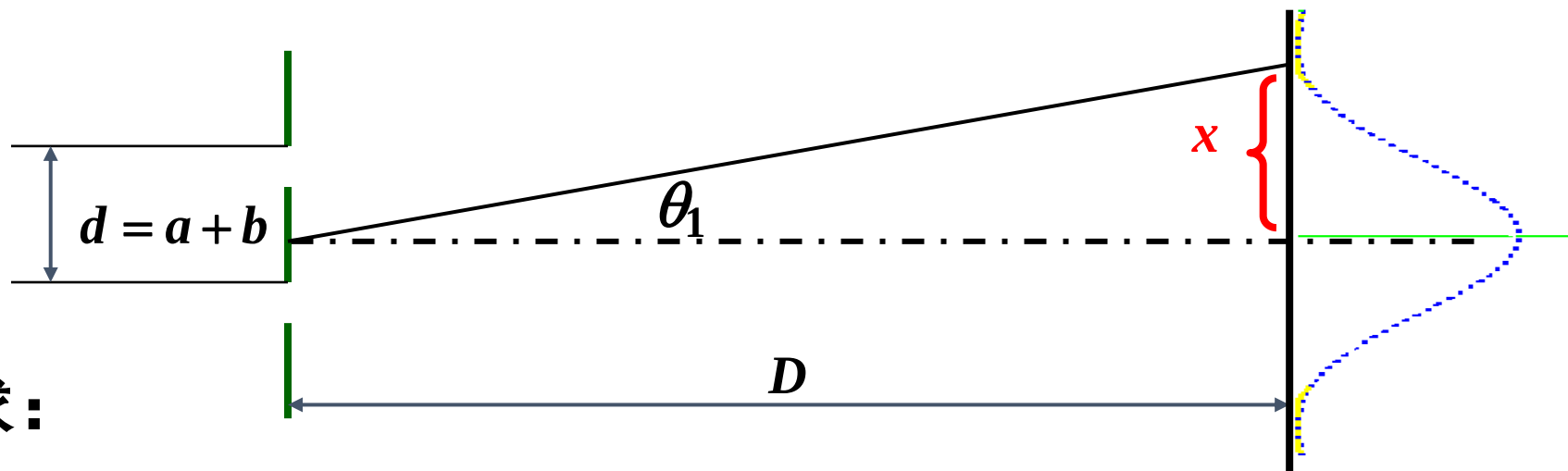
若两束光既满足(3)式,又满足(2)式?
强度为零的衍射光相干
相长干涉的强度仍为零——**缺级**。

缺级条件: $\frac{(2)}{(3)} = \frac{k'}{k} = \frac{a}{d}$ 为整数比时, 缺**k**级

如 $\frac{k'}{k} = \frac{a}{d} = \frac{1}{3}$ 缺第3级明纹, 还缺 6、9、12.....



例. 已知 $D=50\text{cm}$, $\lambda=480\text{nm}$, $d=0.1\text{mm}$, $a=0.02\text{mm}$



求：

- (1) 双缝衍射相邻两条明纹间距
- (2) 中央明纹包络线中的‘x’
- (3) 双缝衍射的第 1 级明纹的相对强度
- (4) 中央明纹的包线中,共包含了几条完整的明纹?
- (5) 中央明纹包线中恰好11条明纹,如何设计 a 、 d ?

解： (1) $\Delta x = \frac{D}{d} \lambda = 2.4\text{mm}$

(2) $x = D \tan \theta_1 \approx D \sin \theta_1 = D \frac{\lambda}{a} = 12\text{mm}$

(4) 中央包线中,共包含了几条完整的明条纹?

包线的第一极小的衍射角

$$a \sin \theta_1 = \lambda \quad \sin \theta_1 = \frac{\lambda}{a}$$

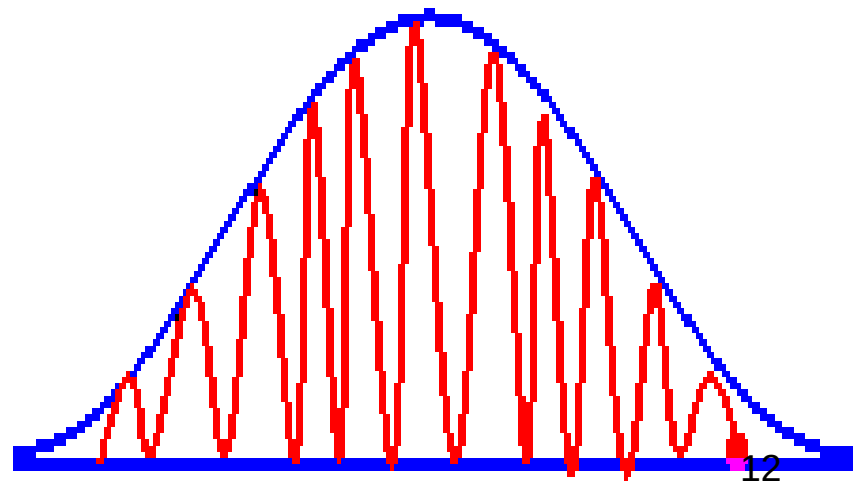
设中央明纹中共有 k 级明纹 $d \sin \theta_1 = k \lambda$

$$d \frac{\lambda}{a} = k \lambda \Rightarrow k = \frac{d}{a} = \frac{0.1}{0.02} = 5 \quad \text{第5级缺级!}$$

包含了 $2 \times 4 + 1 = 9$ 条明条纹

(5) 若要中央明纹的包线中恰好有 11 条明纹, 应如何设计 a 、 d ?

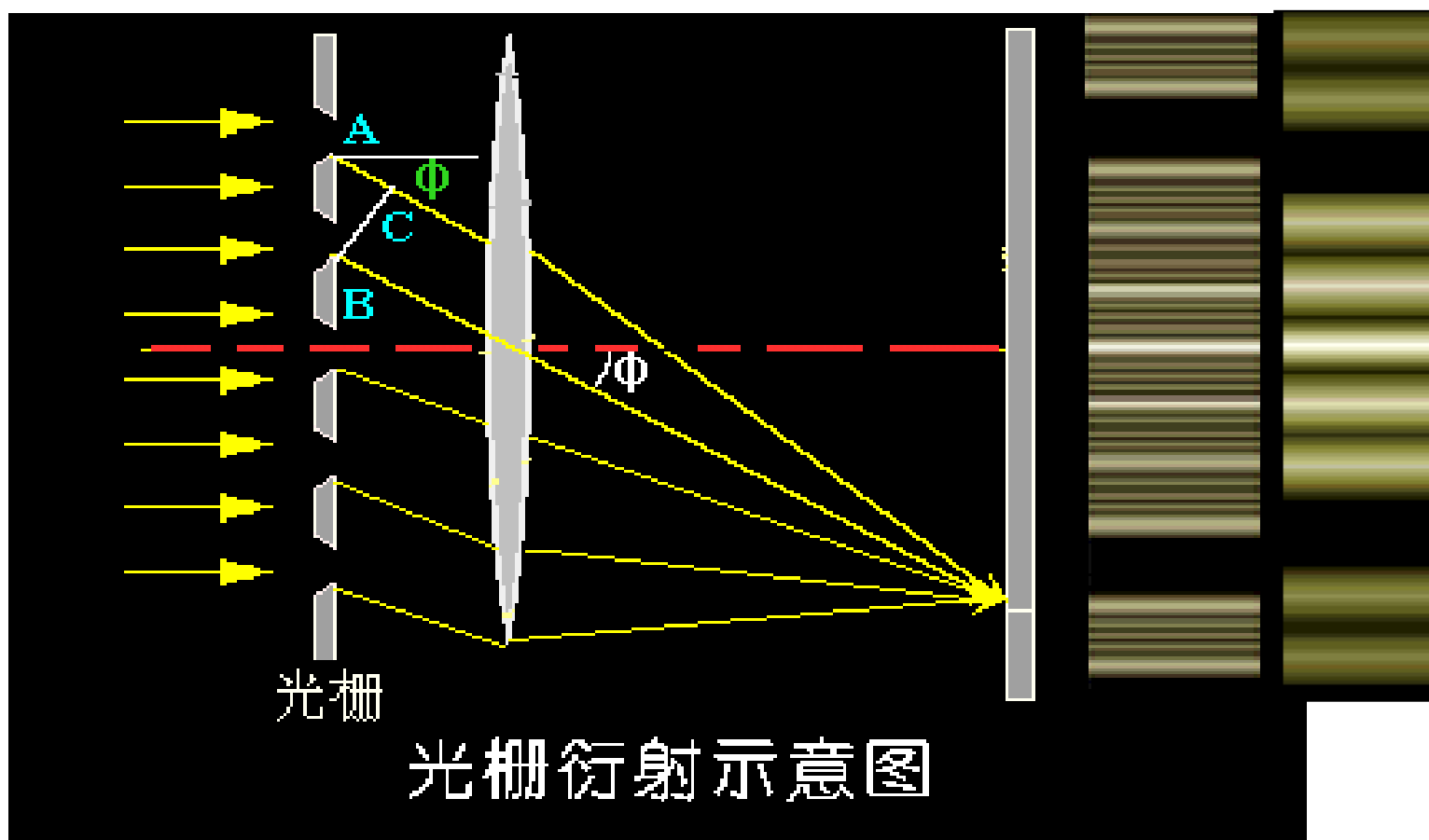
$$\frac{d}{a} = 6$$



四、多缝-衍射光栅

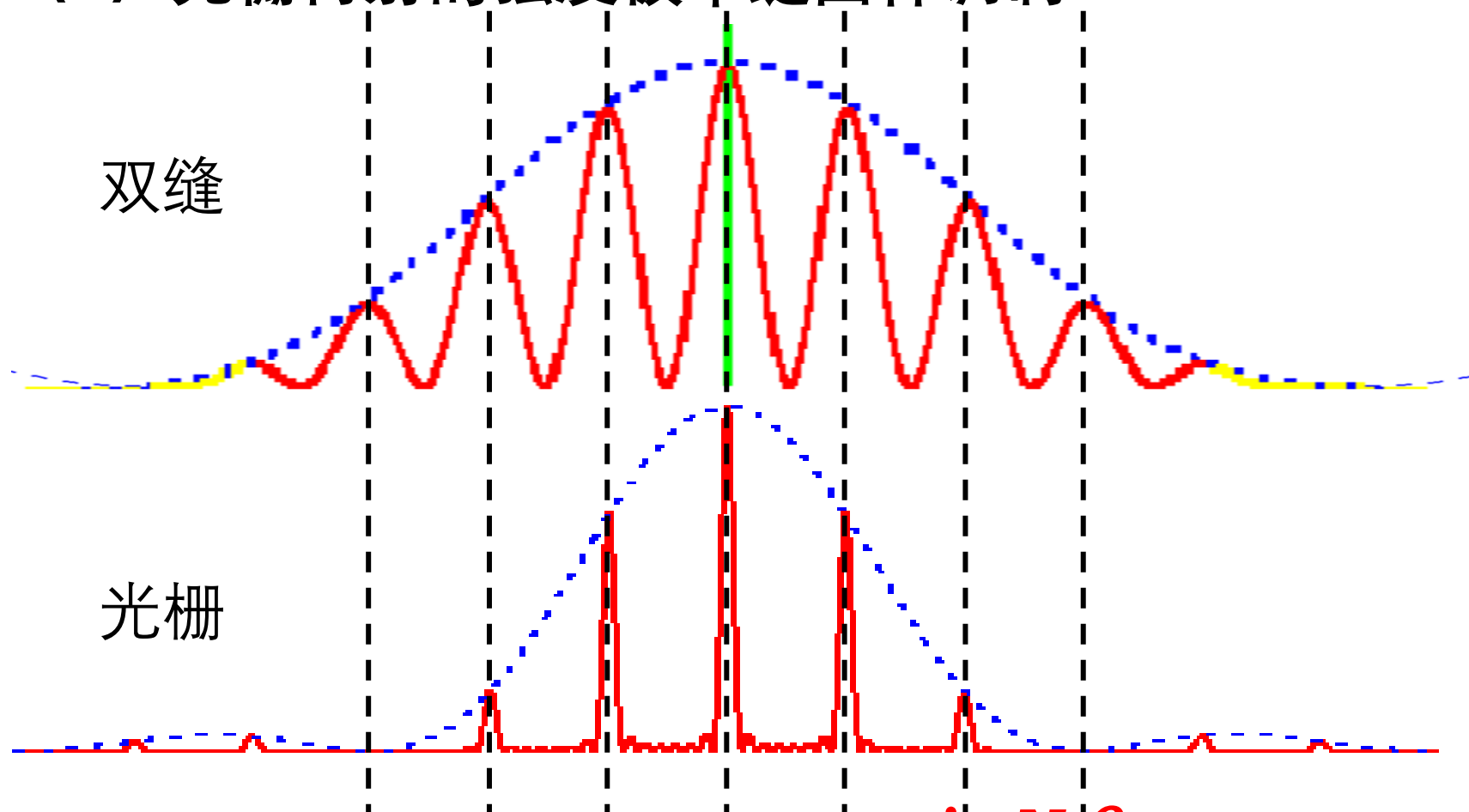
Grating Diffraction

1. 光栅



2.光栅衍射的特点

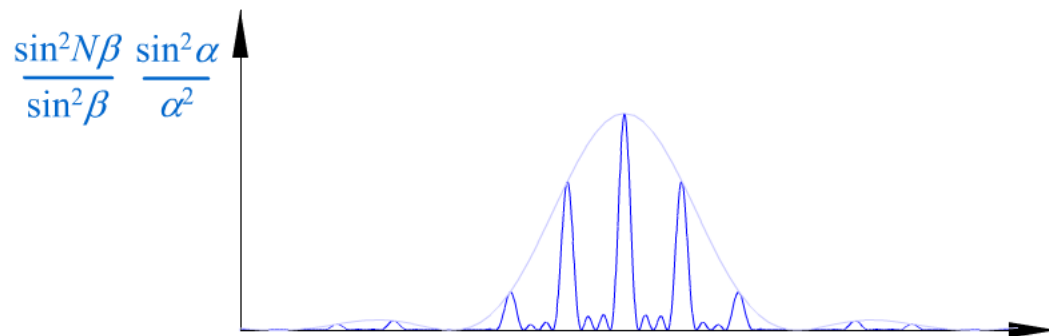
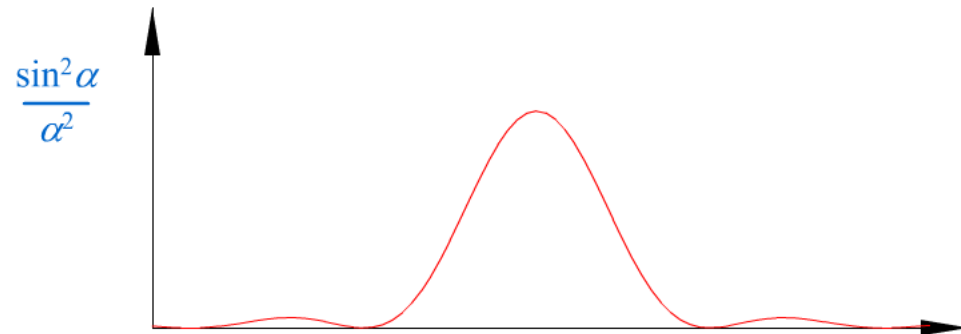
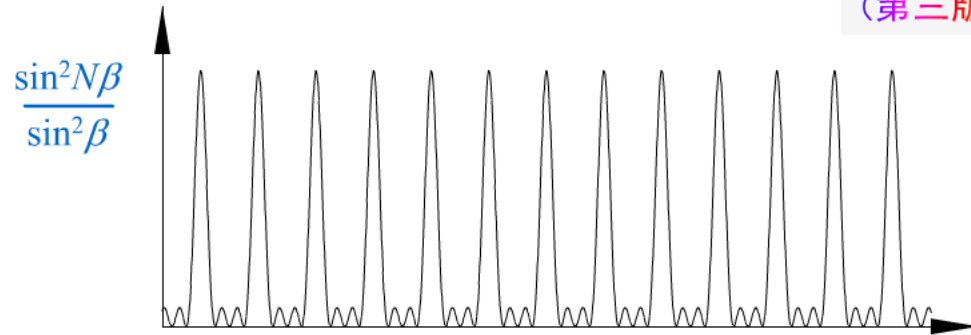
(1) 光栅衍射的强度被单缝图样调制



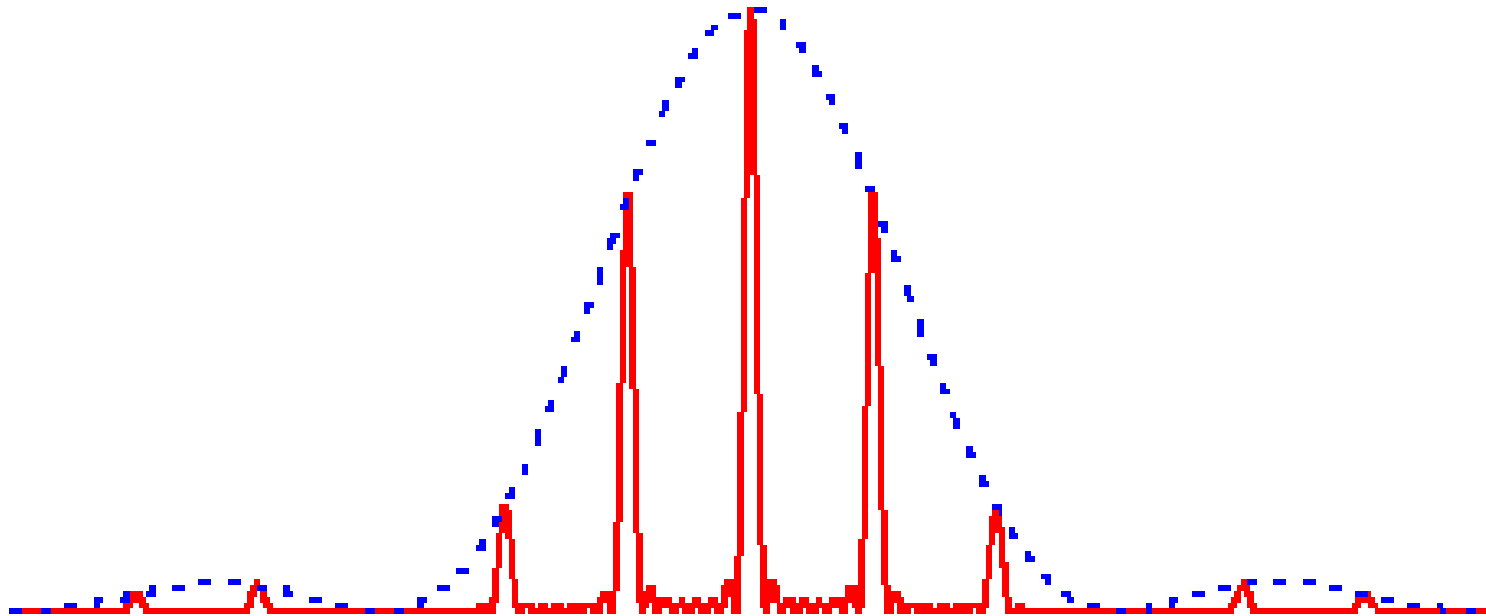
可以证明：
$$I_{\theta} = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N \beta}{\sin \beta} \right)^2$$

$$I_{\theta} = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N \beta}{\sin \beta} \right)^2$$

(第三版)

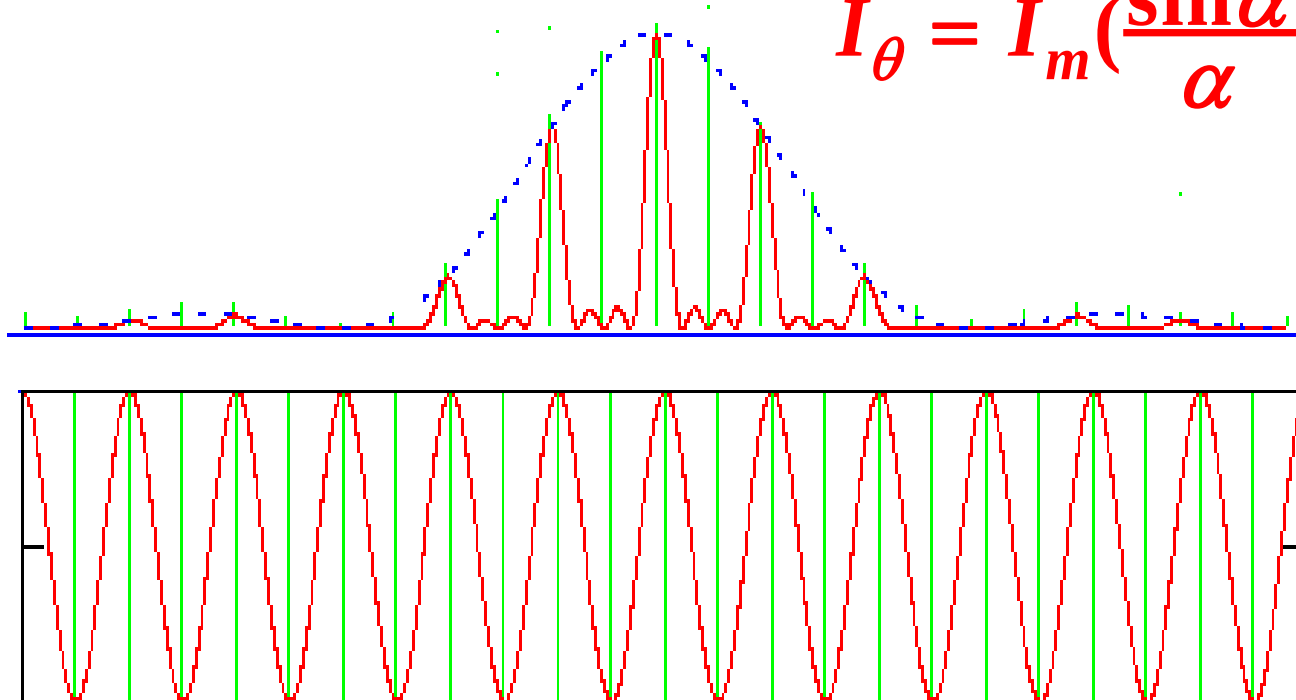


(2) 主极大是明亮纤细的亮纹, 相邻亮纹间是一片宽广的暗区, 暗区中存在一些微弱的明条纹, 称为次极大。



(3) 主极大是各缝出来的衍射光干涉而成的
其位置：

$$I_{\theta} = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N \beta}{\sin \beta} \right)^2$$

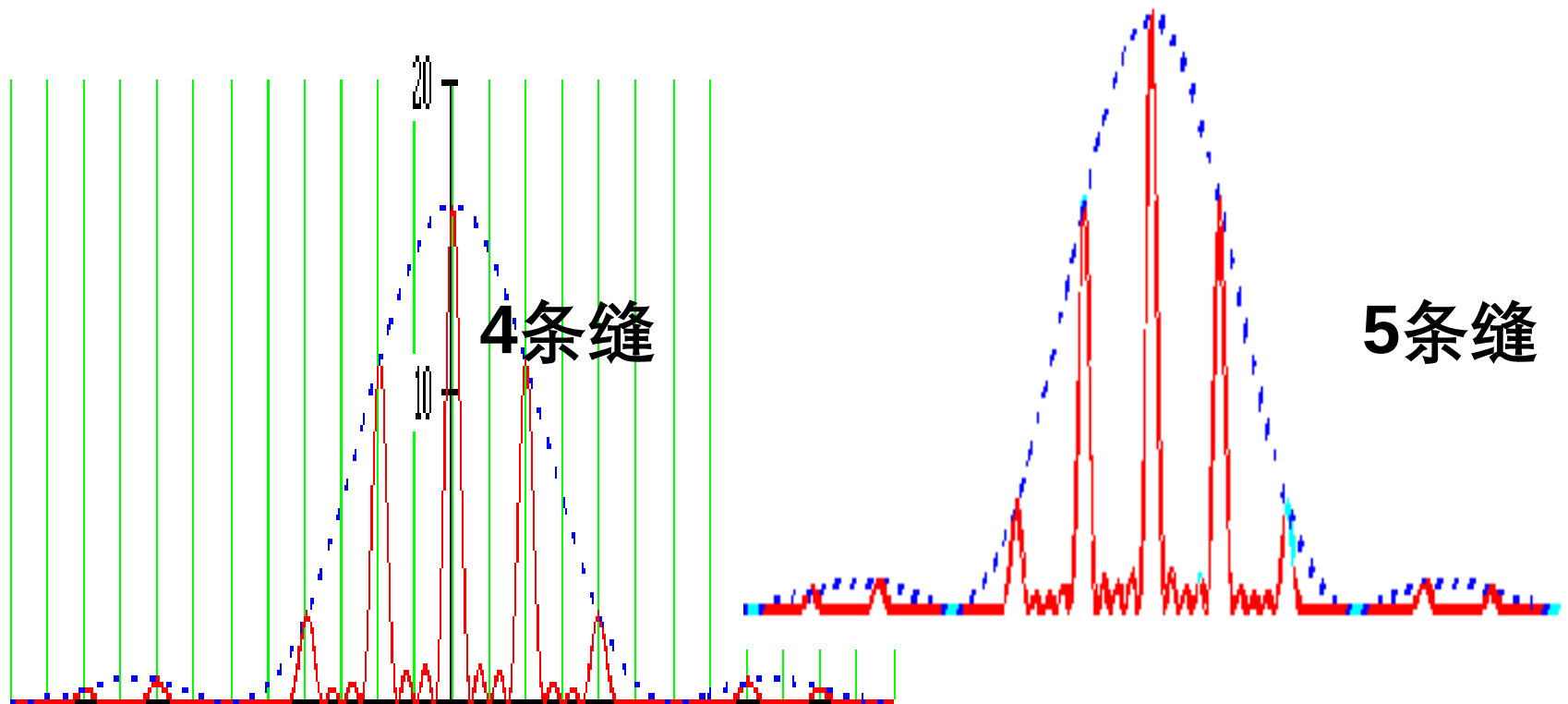


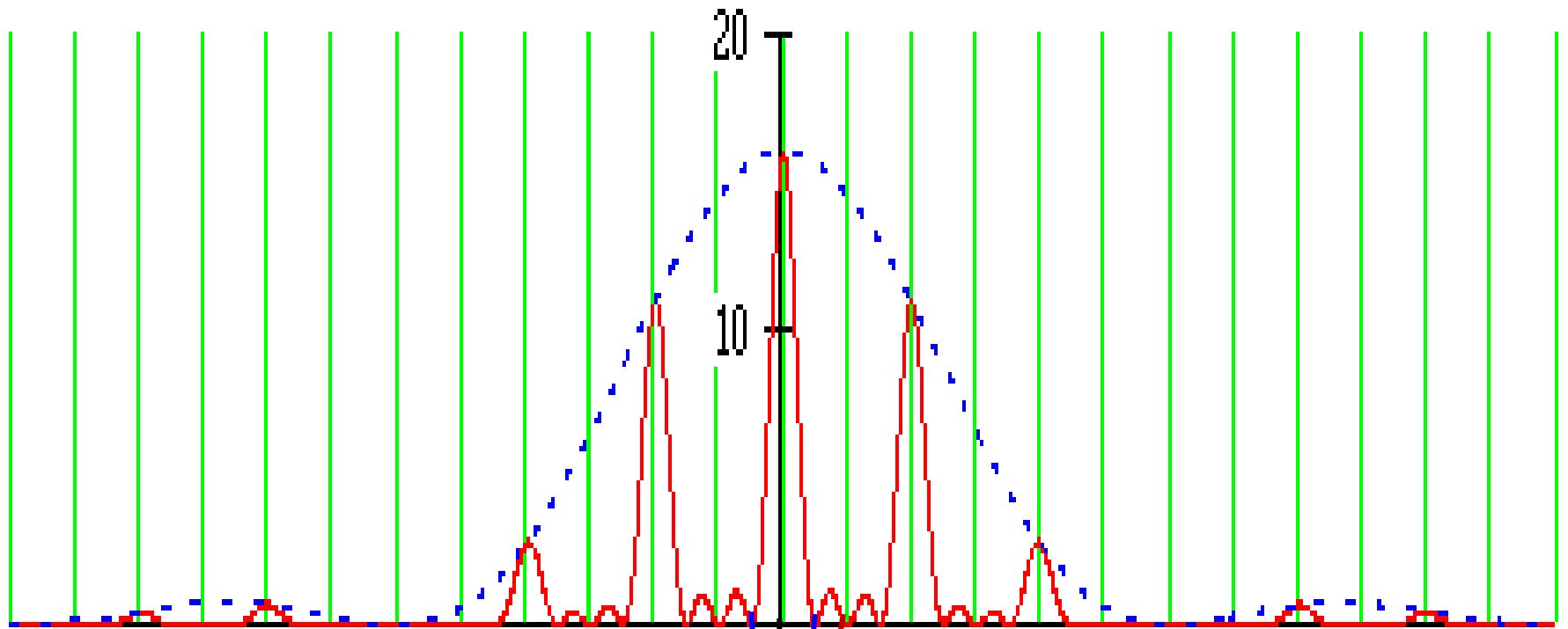
$$\left\{ \begin{array}{l} d \sin \theta = k \lambda \quad (k = 0, \pm 1, \pm 2, \dots) \text{max} \text{ —— 光栅方程} \\ d \sin \theta = \pm k' \left(\frac{\lambda}{N} \right) \quad (k' = 1, 2, \dots, N-1, N+1, \dots, 2N-1, 2N+1, \dots) \text{min} \end{array} \right.$$

N 为缝数, $d = a + b$ 为缝间距 (光栅常数)

$$\left\{ \begin{array}{l} d \sin \theta = k \lambda \quad (k = 0, \pm 1, \pm 2 \cdots) \text{max} \\ d \sin \theta = \pm k' \left(\frac{\lambda}{N} \right) \quad (k' \neq N, 2N, 3N \cdots) \text{min} \end{array} \right.$$

相邻两条明纹间有 $N-1$ 条暗纹



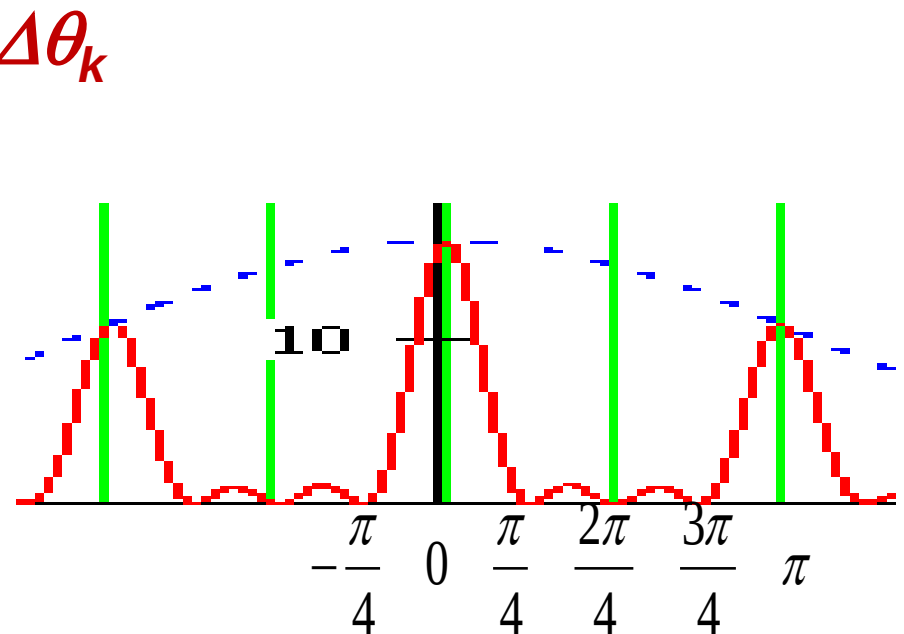


$$I_{\theta} = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N \beta}{\sin \beta} \right)^2$$

if $N = 4$

Max: $\beta = 0, \pi$

Min: $\beta = \frac{1}{4}\pi, \frac{2}{4}\pi, \frac{3}{4}\pi,$



(4) 缺级现象

$\frac{k'}{k} = \frac{a}{d} = \frac{\alpha}{\beta}$ 为整数比时，缺 k 级。



(5) 主极大特别明亮而且尖细，是因为缝多。

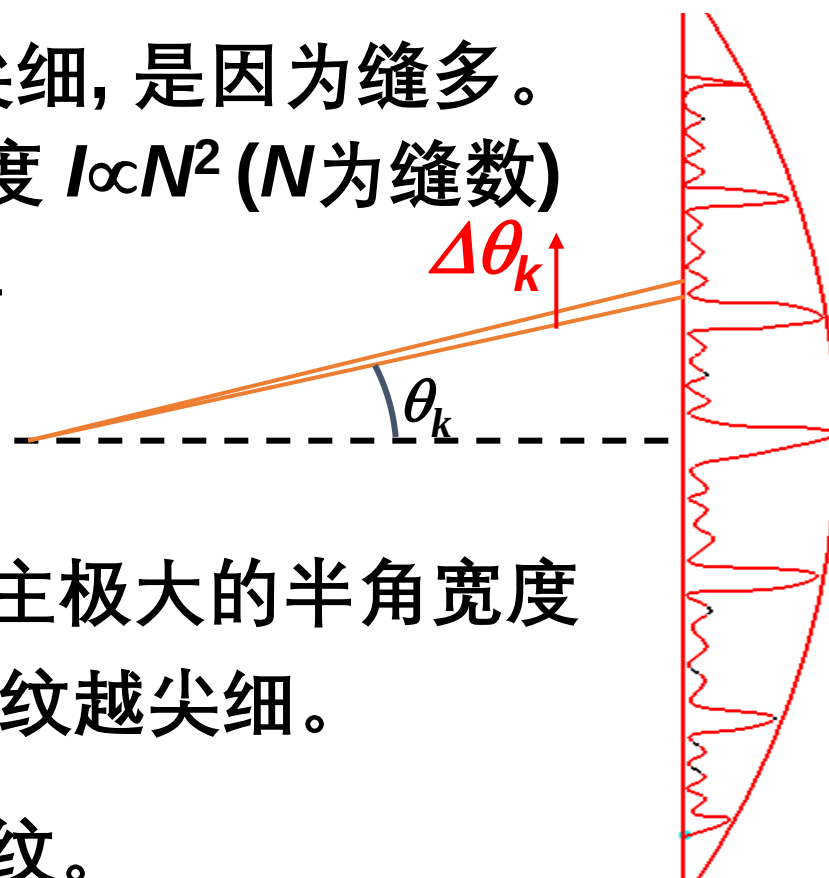
可以证明，主极大强度 $I \propto N^2$ (N 为缝数)

$$\Delta\theta_k \downarrow = \frac{\lambda}{Nd \cos\theta_k}$$

中央主极大 $\Delta\theta_0 = \frac{\lambda}{Nd}$

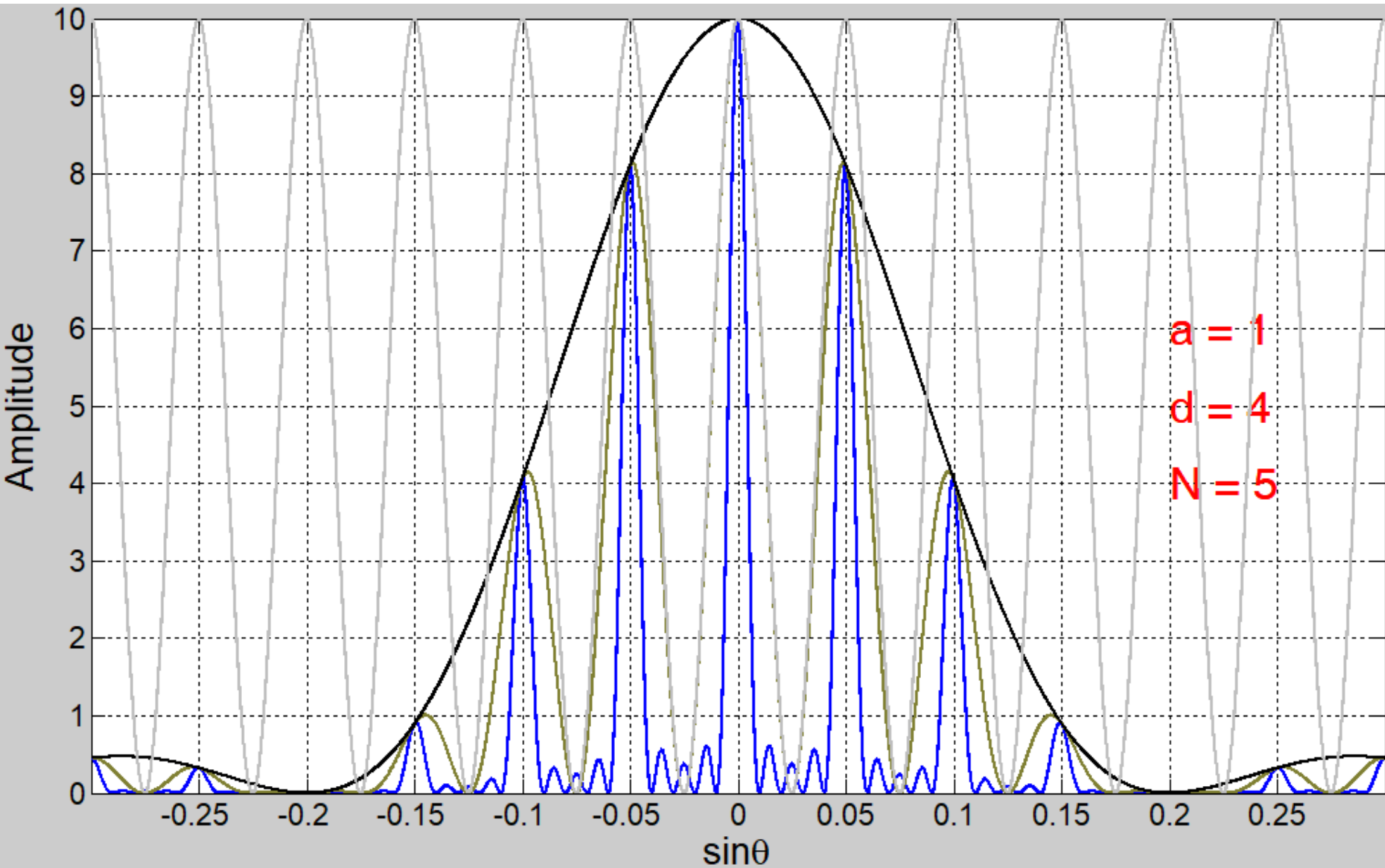
d 为缝间距, $\Delta\theta_k$ 为 k 级主极大的半角宽度

d 一定时，缝数越多，条纹越尖细。



(6) 复色光入射，得彩色条纹。

Matlab simulations

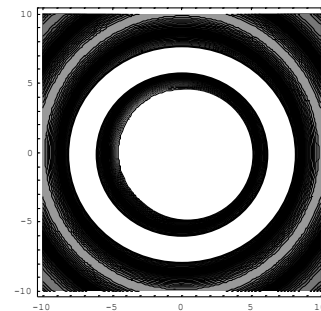
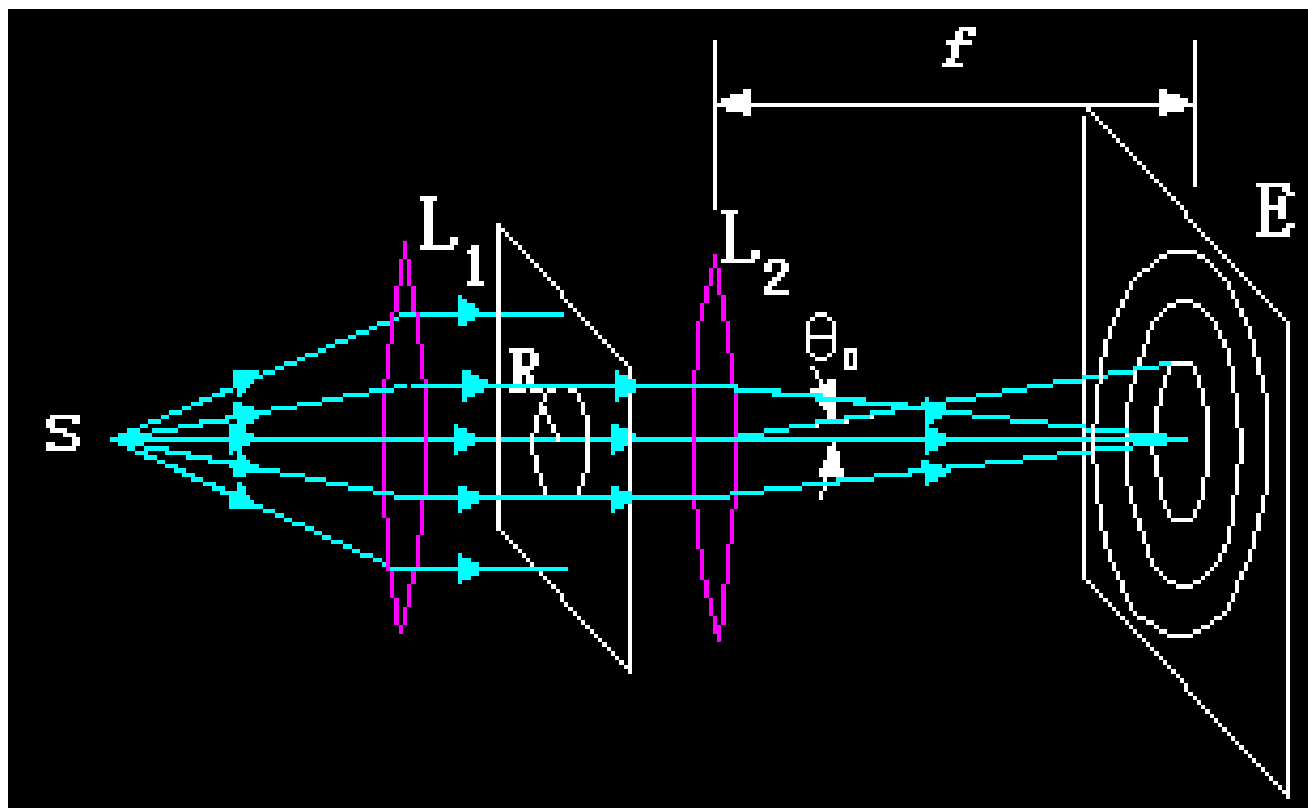


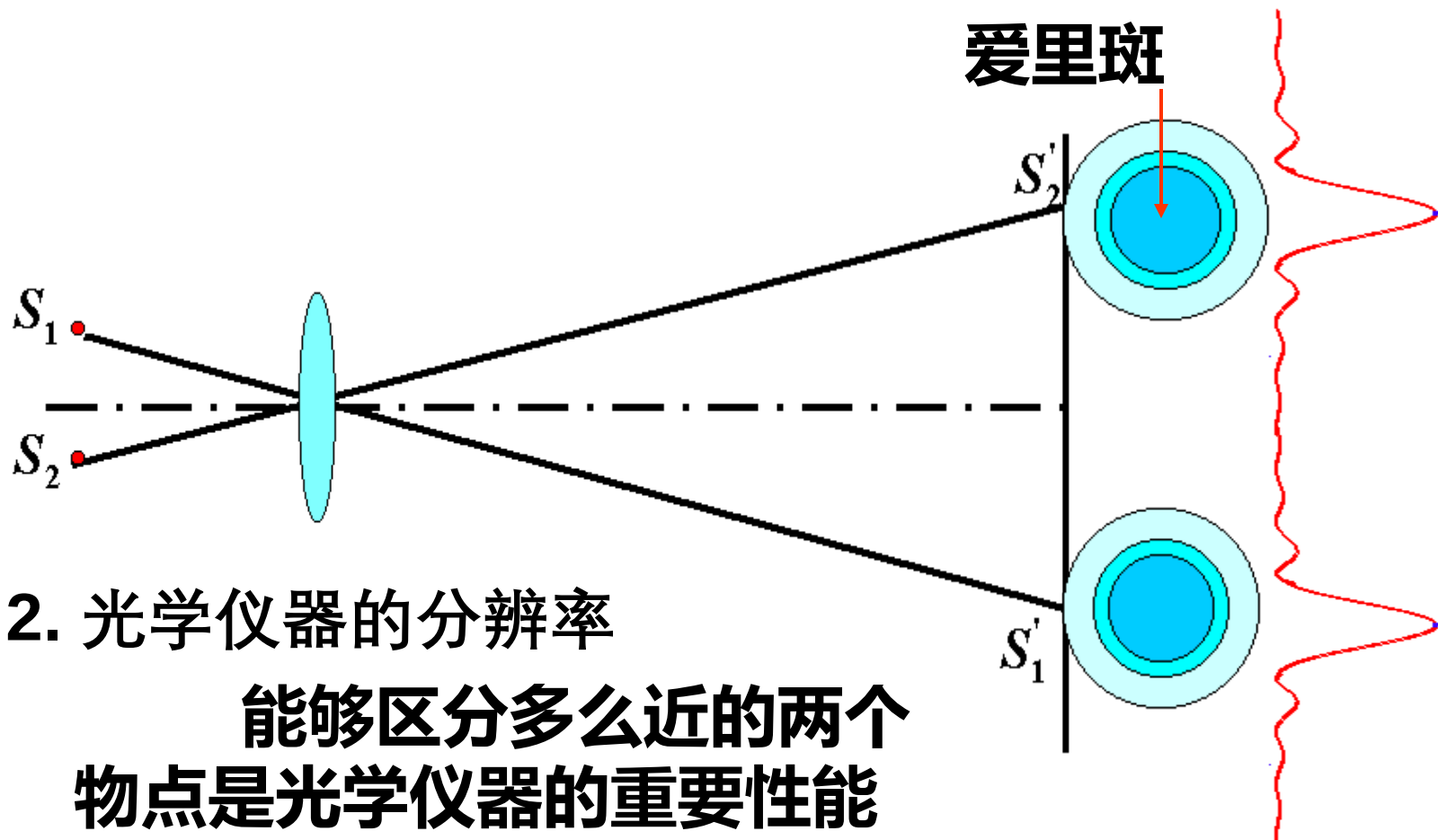
六、圆孔衍射 光学仪器的分辨率

Diffraction by a Circular Aperture

Resolution of Optical Instruments

1. 圆孔夫琅和费衍射

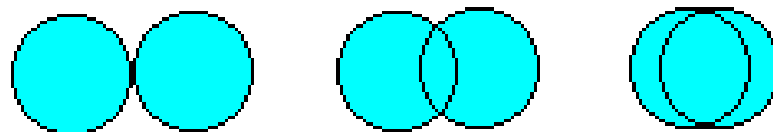




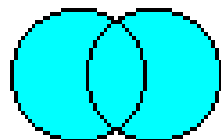
2. 光学仪器的分辨率

能够区分多么近的两个物点是光学仪器的重要性能

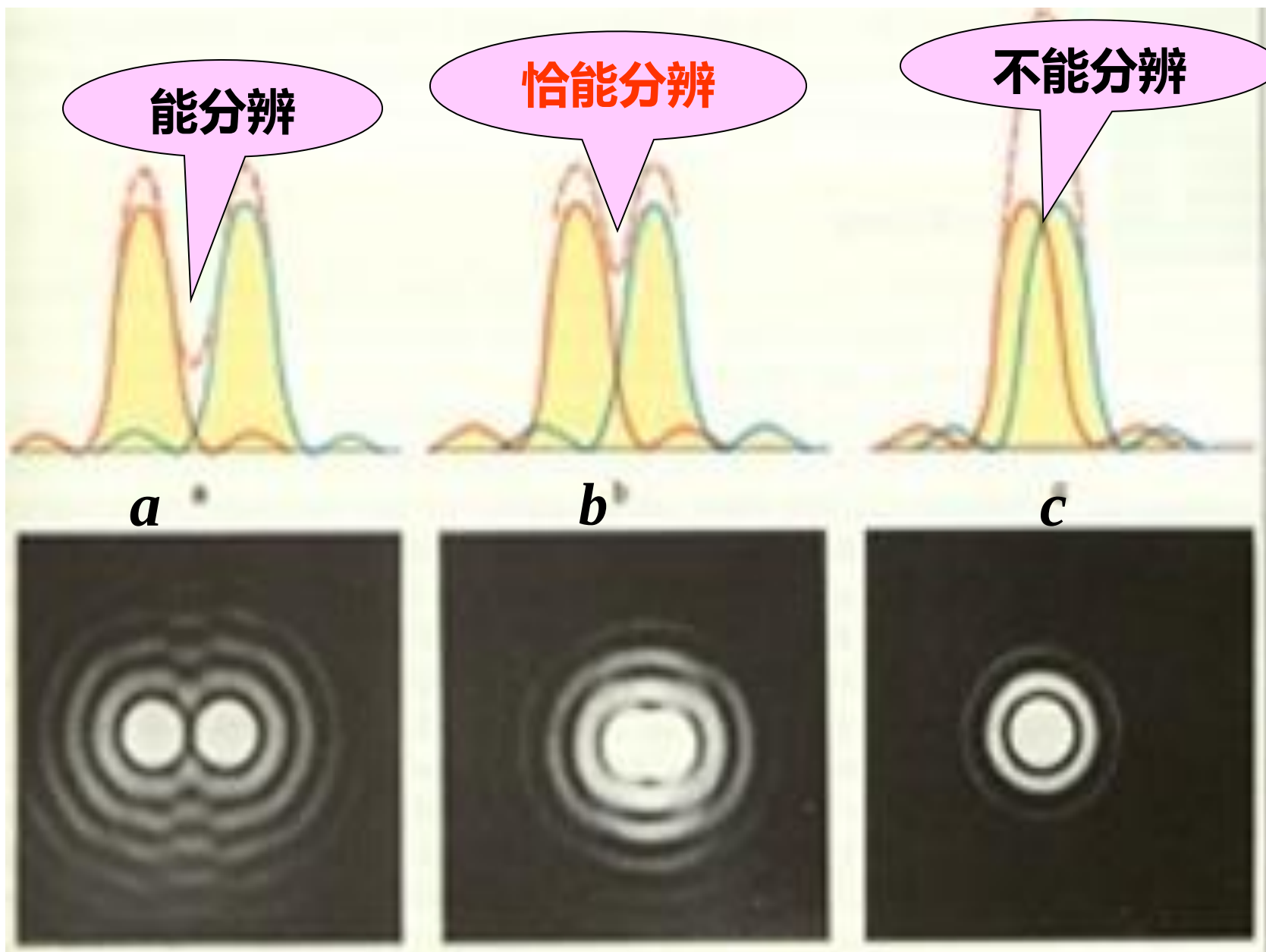
(1) 透镜的分辨率

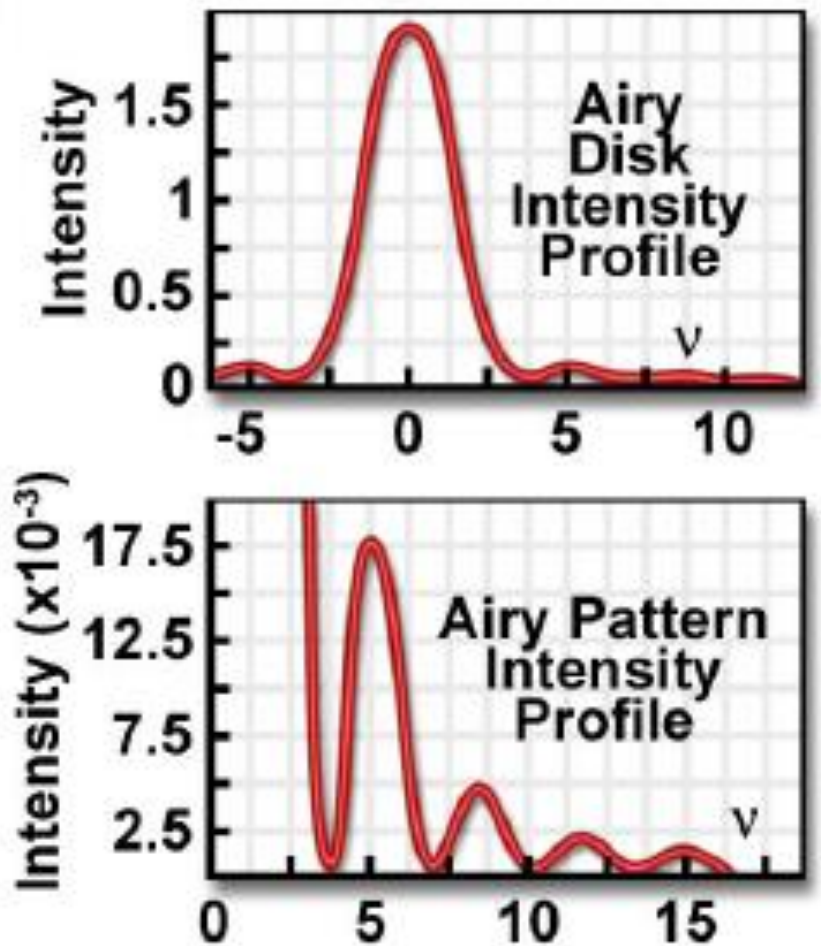
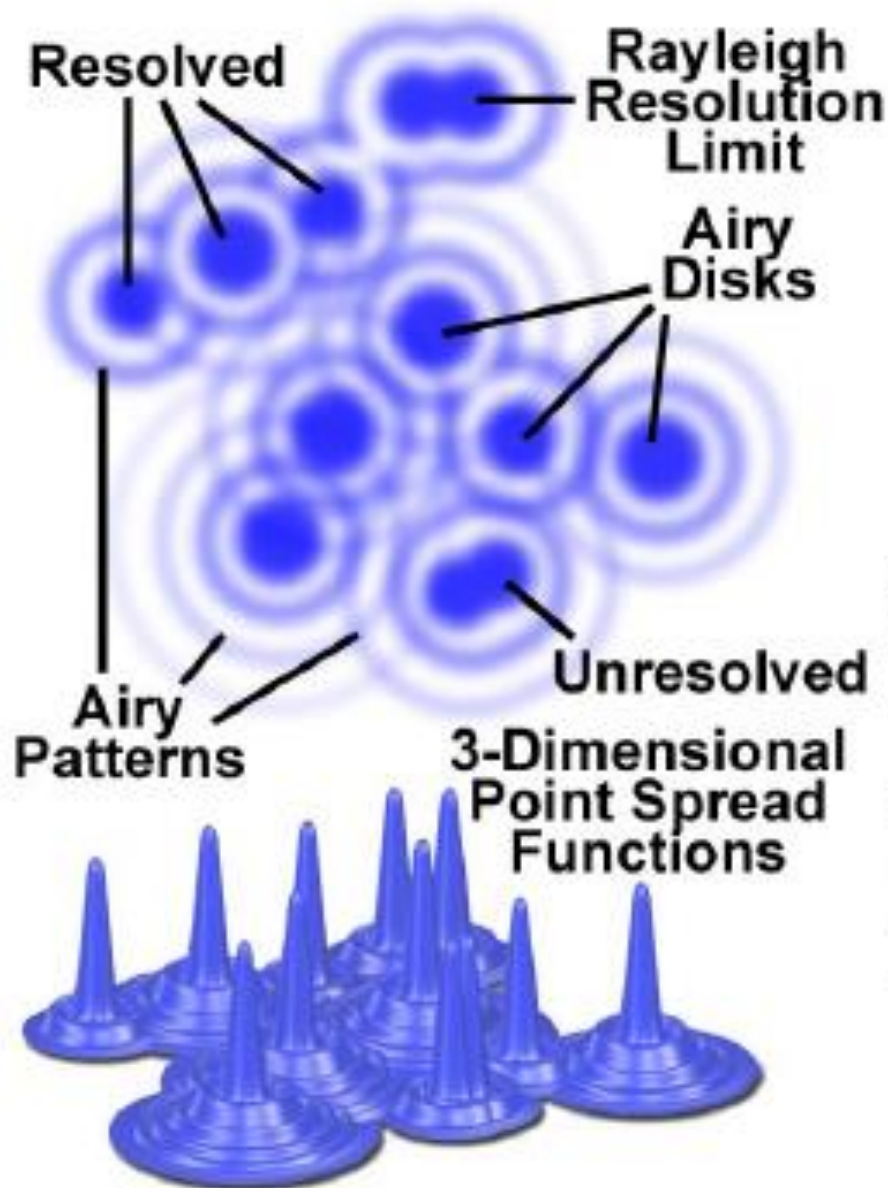


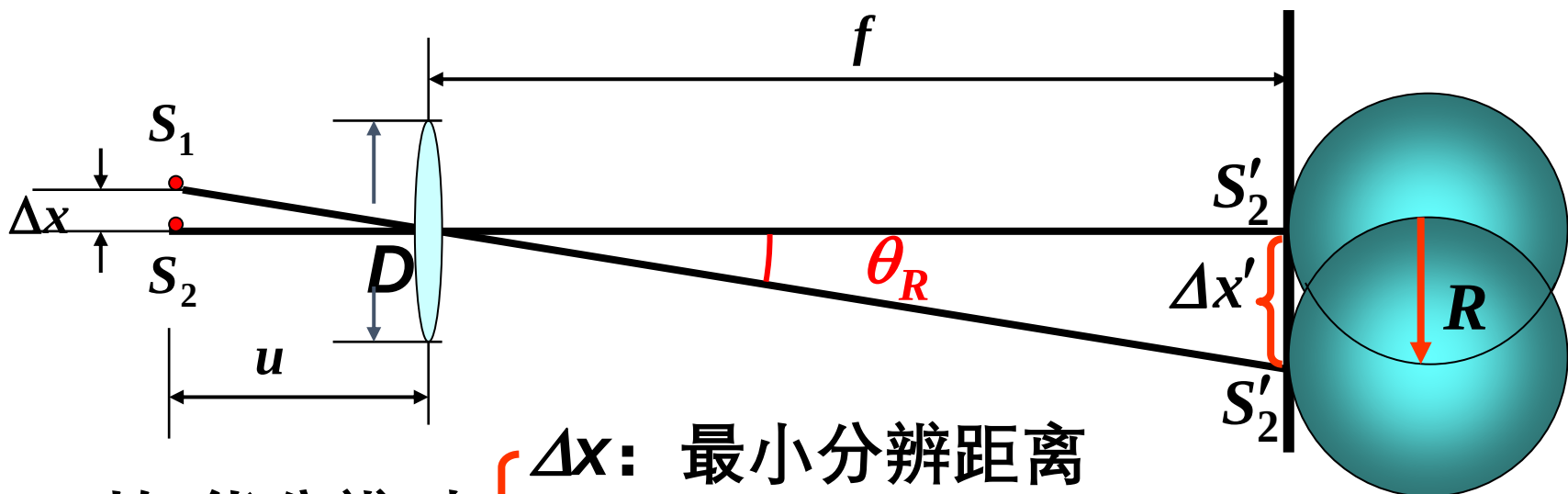
瑞利判据



刚好可以辨别







刚好能分辨时 $\left\{ \begin{array}{l} \Delta x: \text{最小分辨距离} \\ \theta_R: \text{最小分辨角} \end{array} \right.$

理论证明 $\theta_R \approx \sin \theta_R = 1.22 \frac{\lambda}{D}$ 第一极小

光学仪器的分辨率（分辨本领）

定义 $\frac{1}{\theta_R} = \frac{D}{1.22\lambda}$ 越大越好 $\left\{ \begin{array}{l} D \uparrow \\ \lambda \downarrow \end{array} \right.$

最小分辨距离 $\Delta x = u \theta_R$

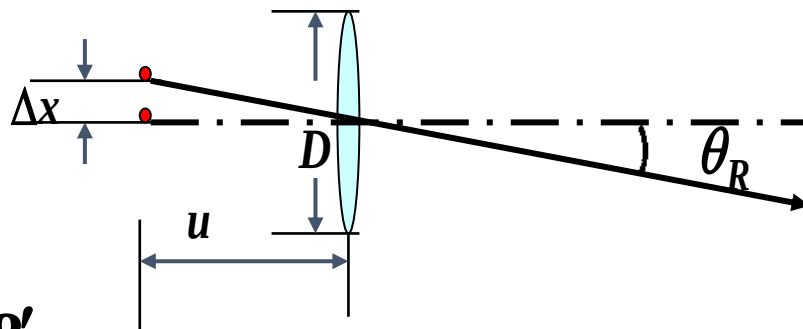
另：像距 $\Delta x' = R = f \theta_R = 1.22 f \frac{\lambda}{D}$

例. 人眼的瞳孔 $D=3\text{mm}$, 用 $\lambda=550\text{nm}$ 的光考虑,
 设 $u=9\text{m}$, $\theta_R=?$ $\Delta x=?$

解: $\theta_R = 1.22 \frac{\lambda}{D}$

$$= 1.22 \times \frac{550 \times 10^{-9}}{3 \times 10^{-3}}$$

$$= 2.23 \times 10^{-4} \text{rad} = 0.8'$$



$$\Delta x = u \theta_R = 9 \times 2.23 \times 10^{-4} \approx 2\text{mm}$$

(2) 光栅的分辨率

光栅、棱镜等类光学仪器的分辨率是指把波长靠得很近的两条谱线 λ_1 、 λ_2 分辨清楚的本领。

$$\Delta \theta_k = \frac{\lambda}{Nd \cos \theta_k}$$

$$\Delta \theta = \frac{k}{d \cos \theta} \Delta \lambda$$

$$\left\{ \begin{array}{l} R = \frac{\lambda}{\Delta \lambda} \\ R = kN \end{array} \right. \Rightarrow \frac{\lambda_1 + \lambda_2}{\lambda_2 - \lambda_1} \quad \Delta \lambda \downarrow, R \uparrow; \quad N \uparrow, R \uparrow$$